Retail Sales Data Analysis Project

Project Overview

This project demonstrates an end-to-end data analysis workflow using Python. A retail sales dataset was cleaned, analyzed, and visualized to generate actionable business insights.

Project Highlights

Cleaned messy data, performed EDA, engineered new features, created business visualizations, and developed KPIs for a Power BI dashboard.

Objectives

Improve data quality, prepare the dataset, perform EDA, identify trends, answer business questions, and communicate findings.

Data Preparation

Handled missing values, standardized categories, converted data types, treated outliers, and created Month, Year, Age Group, Gross Sales, Discount Amount, and Profit Margin features.

Business Questions

Analyzed product performance, discounts, customer segments, regional performance, product profitability, pricing, payment methods, monthly trends, profitability drivers, and Average Order Value.

KPIs

Total Revenue, Total Profit, Total Orders, Total Customers, Average Order Value, Profit Margin, Average Discount, Revenue per Customer.

Visualizations

Bar charts, line charts, scatter plots, box plots, correlation heatmaps, and KPI cards.

Tools

Python, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook, and Power BI.

Skills

Data Cleaning, EDA, Feature Engineering, Statistical Analysis, Data Visualization, Business Analytics, Dashboard Design, and Reporting.

Business Impact

Provided insights into product performance, pricing, customer behavior, regional sales, and profitability to support data-driven decisions.

Conclusion

The project demonstrates the complete analytics lifecycle from raw data to actionable business insights.